Prathyusha Murala

+1 (680) 356-8545 pmurala.work@gmail.com linkedin.com/in/prathyushamurala github.com/prathyusha1231 datawithprats.com

Education

Syracuse University **August 2023 – May 2025**
Master's, Applied Data Science *Syracuse, NY*

Dayananda Sagar University **August 2018 – June 2022**
Bachelor's, Computer Science *Bengaluru, India*

Skills

AI Development: Claude Code, GPT-4, LangChain, LangGraph, Prompt Engineering, RAG, AI Code Review, LLM Evaluation
Full-Stack: Python, FastAPI, JavaScript, SQL, Tailwind CSS, Plotly.js, HTML/CSS, Jinja2, Pydantic, REST APIs
Cloud & DevOps: AWS (S3, Lambda, SageMaker), Docker, Git/GitHub, Uvicorn, CI/CD, Render, ETL Pipelines
Data & Analytics: Pandas, DuckDB, Snowflake, Data Modeling, API Design, Schema Design, pytest, Scikit-learn

Professional Experience

Syracuse University **July 2025 – Present**
Data Science Analyst *Syracuse, NY*

- Built and deployed production RAG system (GPT-4o, LangChain, Pinecone) on AWS serving 60+ daily users; iterated on chunking, embedding, and retrieval pipelines over 2,000+ documents, improving answer accuracy from 65% to 85% through LLM evaluation.
- Developed automated ETL pipelines (AWS Lambda, S3) integrating data from multiple source systems across 10 programs; partnered with cross-functional stakeholders to reduce data latency from 2 weeks to 2 days, driving faster iteration cycles.
- Performed deep-dive analysis on 50K+ user interaction records using Python and SQL; engineered 15+ features and built predictive models (AUC: 0.82) that surfaced behavioral patterns driving product improvements, improving user retention by 10%.
- Defined and owned 8 KPIs with academic advisors and program chairs; delivered weekly insight reports to leadership that directly informed resource allocation and strategic planning decisions.

PricewaterhouseCoopers (PwC) **March 2022 – May 2023**
Data Scientist Associate *Bengaluru, India*

- Designed and deployed BERT-based document classification system for enterprise clients processing 130+ assessments annually; sat with stakeholders to translate business requirements into buildable specs, reducing review time by 87%.
- Implemented semantic search pipeline (SBERT, KeyBERT) on Azure for document matching and retrieval; debugged performance issues across data flow, improving classification accuracy by 35% and delivering to enterprise quality standards.
- Built evaluation framework using LDA topic modeling and BLEU scores to validate AI output quality; achieved 92% consistency across document summaries, establishing criteria for when AI output was production-ready.
- Increased fraud detection recall by 28% using XGBoost on 5M+ transactions; applied SHAP to communicate complex model results to business stakeholders across 10+ client portfolios.
- Established model monitoring pipeline (PSI, KS tests) with automated retraining alerts; provided technical support during critical iteration cycles, maintaining 90%+ precision in production.

Wipro Technologies **June 2021 – August 2021**
Data Analyst Intern *Bengaluru, India*

- Created API-integrated analytics dashboard (FastAPI, Plotly Dash) and automated ETL pipeline for time-series data across 5 regions; delivered working application that improved client forecast accuracy by 30% and informed product planning decisions.

Projects & Publications

Ask Syracuse Data: Full-Stack AI Application | *Claude Code, FastAPI, Tailwind, Plotly.js, DuckDB*

- Built end-to-end web application using Claude Code with FastAPI backend, Tailwind CSS frontend, and Plotly.js visualizations; designed 4 REST API endpoints with rate limiting, input validation, and security middleware.
- Integrated 16 datasets (150K+ records) with 3-path query routing, schema governance, and bias detection; wrote 150+ automated tests and deployed via Docker to production on Render and Hugging Face Spaces.

Multi-Agent AI Copilot | *LangGraph, GPT-4, Function Calling, Autonomous Agents*

- Designed multi-agent system using LangGraph with specialized agents (Search, Analyst, Writer) and OpenAI function calling; reviewed AI-generated outputs, implemented guardrails and prompt controls, achieving 85% relevance on 50 test queries.
- Built autonomous workflow where agents collaborate to research, analyze, and generate campaign-ready content; applied systems thinking to handle edge cases in data flow and ensure reliable end-to-end execution.